

Practice Questions Genetic Screening and Genetic Counseling

1. A 37-year old pregnant woman comes to the physician for a routine checkup at 15 weeks gestation. She is in very good health but asks her physician about prenatal screening. Which of the following prenatal test will allow visualization of the chromosomes of this woman's fetus?
 - A. Chorionic villus sampling (CVS)
 - B. Analysis of cell free DNA
 - C. Amniocentesis
 - D. Detailed Ultrasound
 - E. Fetal blood sampling

2. A 26-year old pregnant woman comes to the physician with a family history of Phenylketonuria (PKU), however the exact genetic defect leading to the disorder in the family has not been determined. Which of the following is the most appropriate postnatal diagnostic technique for disease confirmation?
 - A. PCR-RFLP Analysis
 - B. Tandem Mass Spectrophotometry
 - C. Comparative Genome Microarray
 - D. Native Gel Electrophoresis
 - E. Immuno-reactive Trypsinogen
 - F. ASO
 - G. Comparative genome hybridization

3. A 30-year old pregnant woman comes to the physician because she has a family history of Duchenne muscular dystrophy. This woman is a known carrier for a deletion of exons 18 – 26 of the *DMD* (dystrophin) gene. Which of the following prenatal diagnostic procedures would most likely be used to test this woman's fetus?
 - A. Chorionic villus sampling (CVS)
 - B. Maternal serum AFP screening
 - C. Amniocentesis
 - D. Detailed Ultrasound
 - E. Fetal blood sampling

4. According to the list of criteria for Postnatal or Newborn screening, which of the following best fits the requirements?
 - A. Early intervention is effective and critical
 - B. A treatment does not necessarily have to be available
 - C. The disorder produces irreversible damage
 - D. The reliability of the lab test is not a critical factor
 - E. A genetic diagnostic test must be available

5. A 13-year-old boy is brought to the physician by his parents for a routine examination. There is a positive family history of genetic disorders. Family history shows the patient's maternal grandfather died from Huntington disease at 55 years of age. The patient's paternal grandfather had colon cancer at the age of 35 years and genetic testing showed he was heterozygous for a point mutation in the APC gene leading to a frame-shift in the mature mRNA. Physical and neurological examination is normal. Which of the following is the most appropriate next step for this patient?

- A. Evaluate whether testing of a minor is appropriate
- B. PCR analysis of the triplet repeat sizes of the HTT gene
- C. Next-generation sequencing of the HTT gene
- D. Next-generation sequencing of the APC gene
- E. Standard karyotype analysis

Answers and explanations (Genetic Screening)

1. Answer C. Explanation: Being older than 35 years the woman is at increased risk for chromosome aneuploidies due to non-disjunction events and as such should be offered prenatal screening via CVS or amniocentesis. Due to being 15 weeks gestation, amniocentesis is more appropriate as CVS is offered at 10-14 weeks gestation. Fetal blood sampling is not a screening or diagnostic method as it is more invasive than amniocentesis, or CVS.

Many would argue that non-invasive prenatal screening by analysis of fetal DNA in maternal circulation would be the most appropriate first tier of screening, however, this test does not allow visualization of the chromosomes, it simply can reveal differences in the ratios of the chromosomes.

2. Answer B. Explanation: Postnatal or Newborn screening for PKU involves measuring the levels of phenylalanine in the blood of the newborn via Tandem Mass Spectrophotometry. The sample of blood is obtained via a heel prick 24-48hrs after birth because it is important that the baby has fed before the sample is taken. The blood sample can be screened for numerous disorders at the same time so is a very cost effective procedure.

3. Answer A. Explanation: In this situation where there is a history of a genetic disorder that has a clear diagnosable test and the exact mutation is known as carrier screening has been performed, the most appropriate technique is CVS due to the earlier sampling time but also as it gives more tissue for diagnosis and culturing purposes.

4. Answer A. Explanation: One of the criteria for undertaking neonatal screening for a disorder is that early intervention is effective in controlling the development or preventing the development of symptoms. Note that this is not necessary a cure, it simply implies that an effective treatment must be available. The other criteria are – damage produced by the disorder is irreversible if untreated early in life; a treatment is available for the disorder and a reliable lab test for detection exists.

5. Answer A. Explanation: The disorder is non-life threatening at the patient's current age, (adult onset condition) and can be considered elective at the current time. This consideration includes preventing a range of potential harms including stigma (self-image), discrimination (insurers, employers, teachers, etc.) and also takes away the child's ability to decide for themselves when they reach the age of consent. If no clear benefits exist at the time, it is recommended that parents restrain their desire to know and let the child choose for themselves at the age of consent.